

Question 1: UML Research and System Design

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1 UML research

UML means **Unified Modeling Language**. It is a standard visual language used to specify, visualize, construct and document software systems. A UML class diagram models the static structure of a system: its classes, data, operations and relationships.

A UML class is normally drawn as a rectangle divided into three sections:

1. The top section contains the class name.
2. The middle section contains attributes (data held by objects).
3. The bottom section contains methods (operations objects can perform).

Attributes are written as `visibility name: type`, for example - `balance: float`. Methods are written as `visibility method(parameters): return_type`, for example + `deposit(amount: float): None`.

Visibility symbols are:

- + public: accessible from outside the class.
- # protected: intended for the class and subclasses.
- - private: intended to be hidden inside the defining class.

An association is a structural connection between classes. For example, a Student owns or uses a StudentAccount, while an account records Transactions. Multiplicity describes how many objects can participate in a relationship. 1 means exactly one, 0..1 means optional, and 0..* means zero or many.

2 Design explanation

Student stores identity and academic information. **StudentAccount** controls money, the PIN and the daily spending rule. **Transaction** records an event so the account can keep a history. **CampusService** represents the service used, such as PRINTING or CAFETERIA.

The student name, registration number and programme are public in this simplified educational model because they are ordinary descriptive data. Balance and PIN are controlled through methods/properties; a caller should not freely overwrite them. A transaction belongs to one account and uses one campus service, while one student has one account in this model. An account can have zero or many transactions.

3 Credible references

- Object Management Group. *OMG Unified Modeling Language (UML), Version 2.5.1*. <https://www.omg.org/spec/UML/2.5.1>
- Python Software Foundation. *Classes*. <https://docs.python.org/3/tutorial/classes.html>

The diagram source is in `Smart_Campus_UML.png` and can be rendered with Mermaid-compatible tools.